PLASTIC SHEET AND STRIP (POLYOLEFIN)

(For Use in Ammunition)

1. SCOPE - This specification covers one type of a polyolefin film for use in aerial mines.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards.-The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FEDERAL SPECIFICATIONS

UU-P-268	-	Paper, Kraft, Untreated, Wrapping
PPP-B-636	-	Box, Fiberboard
PPP-D-723	-	Drums, Fiber
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-60	-	Tape, Pressure-sensitive Adhesive, Waterproof, for Packaging
PPP-T-76	-	Tape, Pressure-sensitive Adhesive Paper, (For Carton Sealing)

FEDERAL STANDARDS

Fed Std No. 123 - Marking for Domestic Shipment (Civilian Agencies)
Fed Test Method Std No. 406 - Plastics, Methods of Testing

MILITARY STANDARDS

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

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2.2 Other publications.-The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Publication

D 1505	-	Tentative Method of Test for Measurement of Density of Plastics by the Density-Gradient Technique
D 1238	-	Measuring Flow Rates of Thermoplastics by Extrustion Plastometer

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Penn. 19103.)

Federal Food, Drug, and Cosmetic Act, Additive Amendment

21 Code:	Federal Regulations 121.2510 for Contact with Food

(Application for copies should be addressed to the Department of Health, Education, and Welfare, 330 Independence Ave. S.W., Washington, D.C.)

3. REQUIREMENTS

3.1 Materials.-The material shall be flexible (non-rigid) unsupported homopolymer polyethylene film. The film shall be natural color (essentially colorless). The resin used shall have a minimum density of .919 and a maximum Melt Index of 2.0 when tested as prescribed in 4.3.2 and 4.3.6 respectively.

3.2 Form.-The film shall be furnished in the form of flat, cut sheets or in rolls, as specified. Material in roll form shall be either single thickness or lay flat tubing, as specified.

3.3 Physical characteristics.-The finished sheet or strip shall conform with the physical properties specified in Table I.

3.4 Length and width of rolls and tubing.-Length and width of rolls of film and tubing shall be as specified (see 6.2). The tolerance for widths up to 30 inches shall be plus or minus 3/16 inch. For widths from 30 to 50 inches the tolerance shall be plus or minus 3/8 inch. For widths from 50 to 60 inches the tolerance shall be plus or minus 1/2 inch. For widths from 60 to 120 inches the tolerance

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shall be plus or minus 3/4 inch. For widths greater than 120 inches the tolerance shall be plus or minus 1 inch. Each roll shall contain not more than three pieces and no piece shall be less than 10 feet in length.

3.4.1 Core size.-The film shall be furnished on cores having a length equal to or up to 1/8 inch greater than the width of the film furnished. The inside diameter of the core shall be 3 + 1/8 - 0 inch.

3.5 Length and width of sheets.-Unless otherwise specified, standard size trimmed sheets shall be 20 ± 1/16 inch by 50 ± 1/16 inch.

3.6 Workmanship.-The film shall be uniform in color, texture, finish, and other physical properties and shall be free from gels, pinholes, streaks, tears, holes, cracks, blisters, fish eyes, scratches, mottling, wrinkling, folds, particles or foreign matter, and unmelted material. Edges shall be free of nicks and cuts.

3.7 Physical requirements

TABLE I PHYSICAL REQUIREMENTS

Property	Requirements	Applicable Paragraph
Density, g/cm^3	.916, min	4.3.2
Tensile Strength (min) pounds per sq. in.	Machine Direction 2500 Transverse Direction 2000	4.3.3
Ultimate elongation (percent) min.	Machine Direction 600 Transverse Direction 700	4.3.3
Heat Seal Strength (breaking strength of heat seal, % after 1 hr. water boil)	70 min	4.3.4
Coefficient of variation in breaking, % after 1 hr. water boil	10.0 max	4.3.6
Thickness, mils	4.0 - 0.5	4.3.5
Melt Index	2.0 max.	4.3.6
Percent Extractables (freon 113)	1.6 max.	4.3.7

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection.-Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated. A lot for inspection purposes shall consist of material from the same manufactured process submitted for inspection at one time.

4.2.1 Component and material inspection.-In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.2.2 Inspection of end item.-

4.2.2.1 Visual examination of the end item for defects in appearance, construction, assembly, and general workmanship.-Visual examination of the end item shall be in accordance with the defects set forth below. The inspection level and acceptable quality level (AQL) shall be as specified in 4.2.3. The sample unit shall be one (1) sheet or one (1) running yard in a roll sheet or tubing. The lot size shall be expressed in units of rolls of sheets, tubing, or packages of cut sheets, as applicable.

Examination	Defects
Color, texture, finish	Not uniform. Pinhole, streak, tear, hole, porous, crack, blister, scratches, mottling, wrinkle, fold, foreign matter, fish eye, gel. Edge contains nick or cut.

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4.2.2.2 Dimensional examination of the end item.-Dimensional examination of the end item shall be in accordance with the classification of defects set forth below. The inspection level and AQL shall be as specified in 4.2.3. The sample unit shall be one (1) sheet or one (1) roll, as applicable. The lot size shall be expressed in units of sheets or rolls of film or tubing, as applicable.

Examination	Defects
Roll:	
When specified width is 30 inches or less	Varies from specified width by plus or minus 3/16 inch
When specified width is 30 to 50 inches	Varies from specified width by plus or minus 3/8 inch
When specified width is 50 to 60 inches	Varies from specified width by plus or minus 1/2 inch
When specified width is 60 to 120 inches	Varies from specified width by plus or minus 3/4 inch
When specified width is more than 120 inches	Varies from specified width by plus or minus 1 inch
Length	Any roll less than 99 percent of specified length (may be determined by weight)
	More than 3 splices in roll
	Smallest piece of film in roll less than 10 feet
Sheet:	
Length and width	Varies from specified length and width by plus or minus 1/16 inch
Core:	Core length less than width of film furnished
	Core length more than 1/8 inch greater than width of film
	Inside diameter of core not within specified tolerance

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4.2.2.3 Examination of preparation for delivery.-An examination will be made to determine that packaging, packing and marking comply with the requirements of section 5. Defects will be scored in accordance with the table below. The sample unit shall be one (1) shipping container and contents fully prepared for delivery. The lot size shall be expressed in units of shipping containers contained in the inspection lot. The AQL and inspection level for this examination shall be as specified in 4.2.3.

Examination	Defect
Marking - shipping container, interior, and intermediate containers, as applicable	Incorrect, omitted, illegible; improper size, location, sequence, or method of application
Materials	Nonconforming as to type, grade, finish or other category of the specification
	Required component missing or damaged
Workmanship and preservation	Inadequate application of components, such as: poor closure of container flaps; loose strapping, over-tightened strapping, sharp ends on strapping; inadequate stapling or protruding nails; bulging of container; insufficient dunnage or blocking and supports
	Roll not suitably restrained to prevent unwinding
	Telescoping of Rolls
Contents	Not proper length as specified or marked.

4.2.3 Inspection levels and AQL's for examinations.-The inspection levels for determining sample sizes and the AQL's expressed in defects per 100 units shall be as follows:

For examination in	Inspection levels		AQL
	Sheets	Rolls	
4.2.2.1	I	S-3	2.5
4.2.2.2	S-3	S-3	2.5
4.2.2.3	S-1	S-1	2.5

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4.2.4 Testing of the end item.-The end item shall be tested for applicable characteristics listed in Table II. The sample unit quantity for each shall be two (2) consecutive square yards of material for rolls or the equivalent area for sheets. Disregard a minimum of three turns of film from each end of a roll before collecting the sample. In cutting test specimens, no portion of the sample shall be taken closer than 10 percent of the width from either edge of the film. The lot size shall be expressed in units of rolls or film, tubing, or packages of sheets, as applicable. Failure of one or more sample units to meet individual unit requirements shall be cause for rejection of the lot. For those characteristics applicable to the lot average, there shall be no evidence of failure to meet the requirements as specified.

Lot size (rolls or packages of sheets, as applicable)	Sample size
0 to 50	2
51 to 500	3
501 to 35000	5
35001 and over	8

NOTE: In instances where the supplier's operation is converting only, the tests may be considered that of a component in jumbo roll form purchased from an outside source.

TABLE II INSTRUCTIONS FOR TESTING

Characteristic	Requirements Applicable to Indiv unit	Lot Avg	Number Determinations Per Unit
Density		X	1
Tensile Strength and Elongation		X	see "a"
Heat Seal	X		see "a"
Thickness of Film		X	5
Melt Index	X		1
Percent Extractable	X		1

a Ten (10) machine directions and ten (10) transverse directions taken at random from each sample unit.

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4.3 Test methods

4.3.1 Conditioning.-The conditioning period prior to test shall be 48 hours, the atmospheric conditions, unless otherwise specified, shall be 73.5° ± 2°F. and 50 ± 4 percent relative humidity.

4.3.2 Density.-The density shall be determined in accordance with ASTM D 1505.

4.3.3 Tensile strength and elongation.-The tensile properties of the film shall be determined in accordance with procedure Method B of Method 1013 of Fed. Test Method Std. No. 406, except that all specimens shall be tested at 5 inches per minute rather than as specified therein.

4.3.4 Heat seal strength.-The breaking strength of heat sealed seams shall be determined in accordance with procedure Method B, Method 1013 of Fed. Test Method Std. No. 406. The width of the specimen shall be one inch. The one inch wide specimens shall be superimposed upon each other and seamed with a 1/8 inch wide flat heat seal, 1/4 inch from the edge. The initial jaw separation shall be two inches and the seam shall be equidistant between the two jaws. The heat seal breaking factor shall be determined after boiling in water under atmospheric pressure for one (1) hour. The percent retention of breaking strength shall be the heat seal breaking factor after boiling in water for one hour divided by the material breaking factor unconditioned, reported in percent.

$$\text{The coefficient of variation } (V) = \frac{100\ \sigma}{\overline{X}}$$

where: $\overline{X}$ = average value of the sample
σ = defined in Section 5, paragraph 5.45 of Fed. Std. 406

4.3.5 Thickness.-The thickness shall be determined utilizing a dead weight dial micrometer with a flat anvil of 1/4 inch minimum diameter and a 3/16 inch diameter flat surface on the head of the spindle. The micrometer shall have two ground and lapped circular surfaces, each 0.250 + 0.001 inch in diameter and it shall have a capacity of not less than 0.170 inch. The surfaces shall be parallel to within 0.0001 inch and shall move on an axis perpendicular to themselves. The pressure exerted on the specimen shall be within the limits of 23 and 27 psi. The dial spindle shall be vertical and the

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dial shall be at least two inches in diameter. It shall be continuously graduated to read directly to 0.0001 inch and shall be equipped with a tell-tale hand recording the number of complete revolutions of the large hand. The dial indicator mechanism shall be full jeweled. The micrometer shall be capable of repeating its readings to 0.00005 inch at zero setting or on a steel gage block. The frame of the micrometer shall be of such rigidity that a load of three pounds applied to the dial housing, out of contact with weight or the presser foot spindle, will produce a deflection of the frame not greater than 0.0001 inch, as indicated on the micrometer dial.

4.3.5.1 Procedure.-Five specimens, at least 2 by 2 inches in size taken uniformly across the width of the roll, shall be tested. The micrometer shall be placed on a solid level table, free from excessive vibrations. The specimen shall be placed between the micrometer surfaces and the presser foot lowered onto the specimen at a location outside the area to be measured. The presser foot shall then be raised for a distance of 0.3 to 0.4 mils and the specimen moved to the measurement position. The presser foot shall then be dropped onto the specimen. In making a measurement, all points on the peripheries of the micrometer surfaces shall be at least 1/4 inch from the edges of the specimen. Reading of the dial indicator shall be deferred until the presser foot has been been supported by the specimen for at least two seconds or until the micrometer hand becomes stationary. The surfaces of the anvil and spindle head and of the specimen shall be clean and dry. The total load applied by the spindle shall be 4 ounces. One measurement shall be made on each specimen. The measurements of all the specimens of a sample shall be averaged to obtain the thickness of the sample.

4.3.6 Melt index.-Determine the melt index in accordance with ASTM-D-1238, Condition E, reported in grams per 10 minutes.

4.3.7 Percent extractables.-Weigh a strip of the material (4 inches by 8 inches) on an analytical balance. Place weighed sample into a "pipe bomb" (or other suitable pressure bomb) with the vessel three-quarters full of trichlorotrifluoroethane (Freon 113 or equivalent). The sample should be completely immersed in the Freon. Seal the bomb tightly and keep in an oven maintained at 160 ± 2°F. for a minimum of four days. Remove sample from the bomb and dry the material for two (2) hours at 160°F ± 5°F. Allow the material to cool to room temperature and reweigh. Calculate the percent extractables as follows:

$$\text{Percent extractables} = \frac{A-B}{A} 100$$

Where: A = original weight of sample, gms.
B = final weight of sample, gms.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.-Unless otherwise specified in the contract or purchase order, the polyolefin shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.-

5.2.1 Level C.-Polyolefin film, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carrier applicable to the mode of transportation.

5.3 Marking.-In addition to any special markings required by the contract or purchase order, containers shall be marked in accordance with MIL-STD-129. Marking shall include, but shall not be limited to, the following information:

a. Manufacturer's name
b. Product designation
c. Lot number
d. Date of manufacture
e. Number of this specification

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, date of this document and thickness of film required.

6.2 Intended use.-The plastic described in this specification is intended for use in ammunition items.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No.: 1345-A-254

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